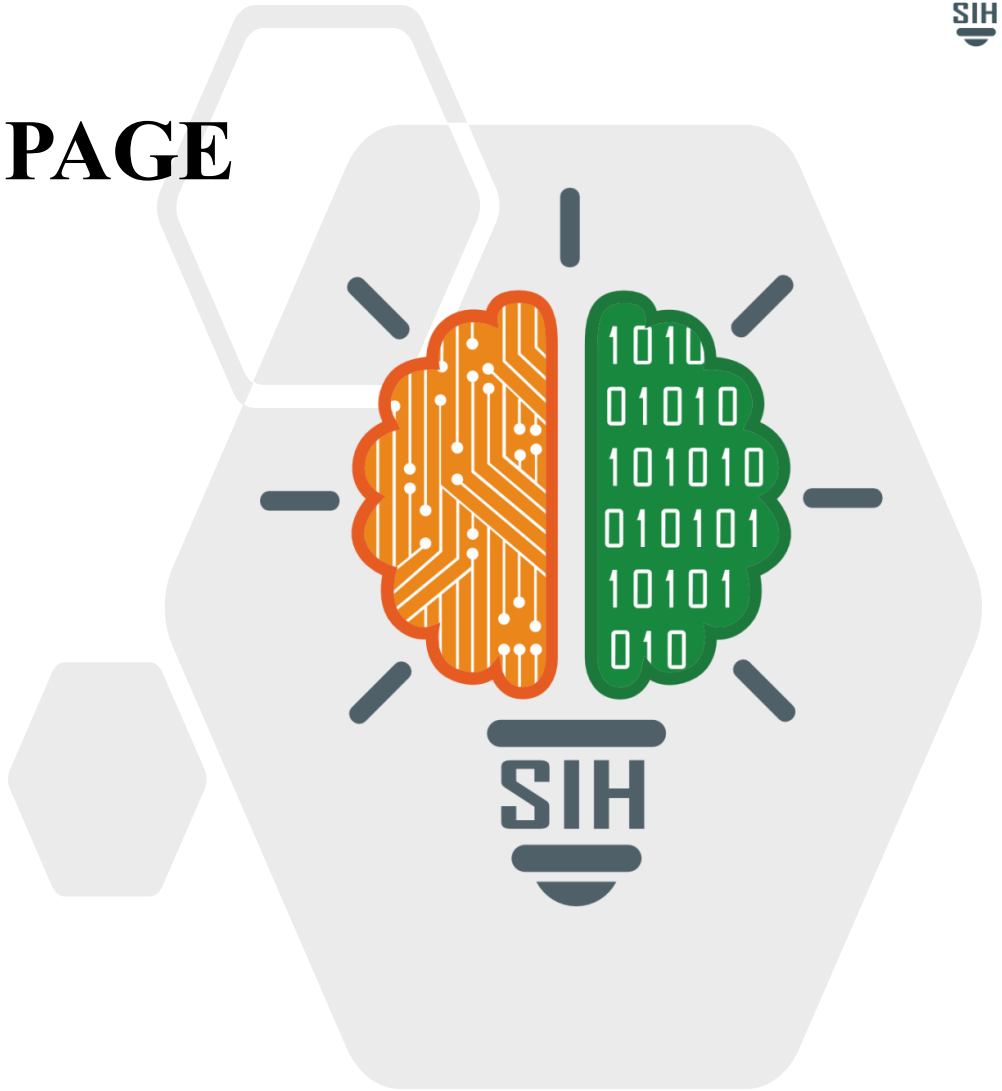
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TITLE PAGE

- **Problem Statement ID –** 26139
- **Problem Statement Title –** Hybrid Quantum Machine Learning Platform for Early Disease Detection
- **Theme –** MedTech / BioTech / HealthTech
- **PS Category –** Software
- **Team ID –** 176689
- **Team Name –** Trace Force_A0



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QuantaDx: Hybrid Quantum–Classical AI for Early Disease Detection

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❖ Proposed Solution

Early detection · reduce missed cases · explain predictions

THE PROBLEM WE SOLVE

Cancer, cardiovascular & neurological disease is often caught late; biomedical data (EHR, genomics, imaging) is **high-dimensional, noisy & imbalanced** → **missed positives**, **black-box** models; near-term **NISQ** devices are small & noisy.

- 43 M NCD deaths / yr [16]
- 14.6 L cancer cases, India (2022) [17]
- Breast ca. 5-yr: ~100%→34% [18]

OUR SOLUTION — QuantaDx

PoC · 5-fold CV · UCI WDBC

Classical pre-processing & feature engineering → **MI/LASSO → PCA 30 → 8** → **ZZ-map quantum kernel on 8 qubits** fused with an RBF kernel (**hybrid Q+C kernel**) → **sensitivity-first threshold** → explainable risk score.

- **96.7%** sensitivity
- **98.0%** specificity
- **7 / 212** cancers missed

WHY Hybrid Quantum ML?

- **Why quantum:** ZZ-map quantum kernel uses superposition & entanglement to build an alternative feature-space similarity that may capture structure classical kernels miss [3, 4]
- **Why hybrid:** CPU cleans & compresses for small NISQ QPUs; $K = \alpha \cdot K_Q + (1-\alpha) \cdot K_{RBF}$ fuses complementary similarities
- **Why sensitivity-first:** threshold set for ≥95% sensitivity → fewer false negatives in screening support
- **Why not only XGBoost?** Same 5-fold CV — **Hybrid Q+C: 7 missed vs XGBoost: 12** (McNemar p = 0.004)

HOW It Works — 6-Step Hybrid Pipeline

CPU · classical + 8-qubit QPU → Hybrid Q+C → Decision support

1. **Data Ingestion** — EHR · lab data · CSV · PHI checks
2. **Pre-processing** — KNN impute · IQR · SMOTE (train folds only)
3. **Feature Reduction** — MI/LASSO selection → PCA: 30 → 8 (93% variance)
4. **Quantum Encoding** — ZZ feature map → entangled 8-qubit state
5. **Hybrid Kernel SVM** — $K = \alpha \cdot K_Q + (1-\alpha) \cdot K_{RBF}$ · inner-CV tuning; VQC / QNN benchmarked
6. **Calibrate & Explain** — ≥95% sensitivity threshold · risk tier · SHAP

How It Addresses the Problem

- **High-dim data** → feature engineering · MI/LASSO selection · PCA → 8 qubits
- **Noisy / imbalanced** → cleaning · KNN imputation · IQR clip · SMOTE in train folds
- **Black-box AI** → SHAP on full pipeline → worst radius, concave points
- **Missed diagnoses** → ≥95% sensitivity threshold: only 7 / 212 missed

Innovation & Uniqueness

- Hybrid quantum + classical kernel (α-mix)
- Bandwidth tuning mitigates kernel concentration
- Sensitivity-first screening threshold
- Honest benchmarks: k-fold CV + McNemar
- Interpretable: SHAP on clinical features
- Scalable & hardware-agnostic: Qiskit · PennyLane · IBM

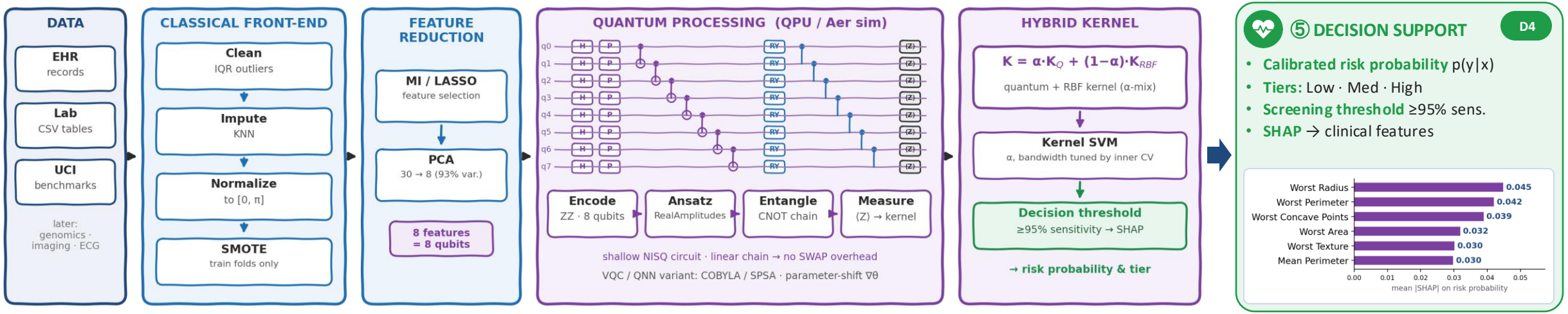
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TECHNICAL APPROACH

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❖ Methodology & Process for Implementation

CPU compresses → QPU learns → CPU decides & explains



⑥ PLATFORM: React + Vite dashboard ⇄ FastAPI (REST + WebSocket, auth & validation) ⇄ Celery + Redis async jobs ⇄ ML inference (quantum + RBF kernels) ⇄ PostgreSQL + MLflow registry · Docker · CI/CD

❖ Technologies to be Used

QUANTUM	Qiskit	PennyLane	IBM QPU	Aer Sim	Runtime	ZNE
AI / ML	Python	sklearn	PyTorch	XGBoost	SHAP	Pandas
BACKEND	FastAPI	Celery	Redis	Postgres	MLflow	Docker
FRONTEND	React	TS	Vite	Tailwind	Plotly	CI/CD

❖ Working Prototype — Screening Support



Training & evaluation dashboard · result visualization · UCI WDBC PoC

LIVE DEMO
quantadx.vercel.app
Screening-support web app

DEMO VIDEO
youtu.be/jIR71ZNFoB8
Video walkthrough of prototype

GITHUB REPOSITORY
github.com/KoduruChandrasekhar/QuantaDx
Code · documentation · setup guide

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FEASIBILITY AND VIABILITY

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❖ Analysis of the Feasibility

Technical ●●●●●

Open-source Qiskit & PennyLane [12, 13]; 8-qubit, shallow NISQ-ready kernels; PoC done

Data ●●●●○

Public benchmark first (UCI) → external validation; hospital data via MoU

Hardware ●●●●○

Aer simulator-first → real-QPU validation; noise-aware eval, ZNE / readout mitigation

Economic ●●●●●

Open-source software stack; simulator-first PoC; no licence cost

Operational ●●●●○

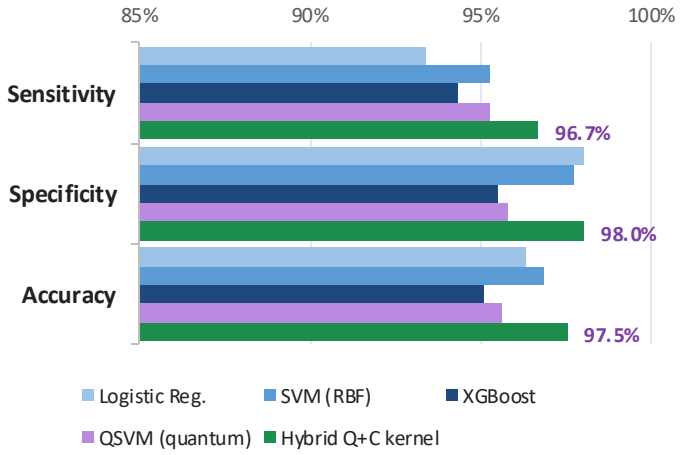
Web app + Docker; clinician-friendly workflow, no quantum expertise required

Regulatory ●●●○○

DPDP Act 2023, ICMR AI-ethics; de-identification, access control; decision support, not diagnosis

❖ Proof of Concept — Screening / Diagnostic Support

UCI Breast Cancer Wisconsin (Diagnostic), n = 569 (212 malignant) [14] · PCA → 8 qubits · ≥95% sensitivity threshold set on inner folds · 5-fold cross-validation · classical baselines: LR, SVM, XGBoost, QSVM



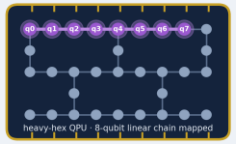
MISSED CANCERS (of 212) · lower is better

Logistic Reg.	14
SVM (RBF)	10
XGBoost	12
QSVM	10
Hybrid Q+C	7 ✓

McNemar (paired CV predictions): significant vs XGBoost (p = 0.004); vs LR (p = 0.12) and SVM (p = 0.22) not significant.

❖ Challenges & Risks → Strategies to Overcome

CHALLENGE / RISK	RISK	MITIGATION STRATEGY
Barren plateaus	High	Shallow hardware-efficient ansatz, local cost; kernels need no PQC training
NISQ noise & decoherence	High	Noise-aware Aer training, readout mitigation, ZNE, SPSA
Few qubits vs high-dim data	Med	MI / LASSO + PCA to qubit budget, data re-uploading
Kernel concentration	Med	Bandwidth tuning + α-mix with RBF, chosen by inner CV
Privacy & data leakage	High	De-identification, DPDP 2023; SMOTE / scaling fit on train folds only
XAI cost on quantum models	Med	KernelSHAP on cached states: 360k evals in 26 s (computational efficiency)



NISQ-ready by design

8 qubits · shallow ZZ-map circuits · linear chain → connected heavy-hex path, no SWAP overhead · readout mitigation, ZNE, SPSA/COBYLA. Simulator-first → real-QPU validation.

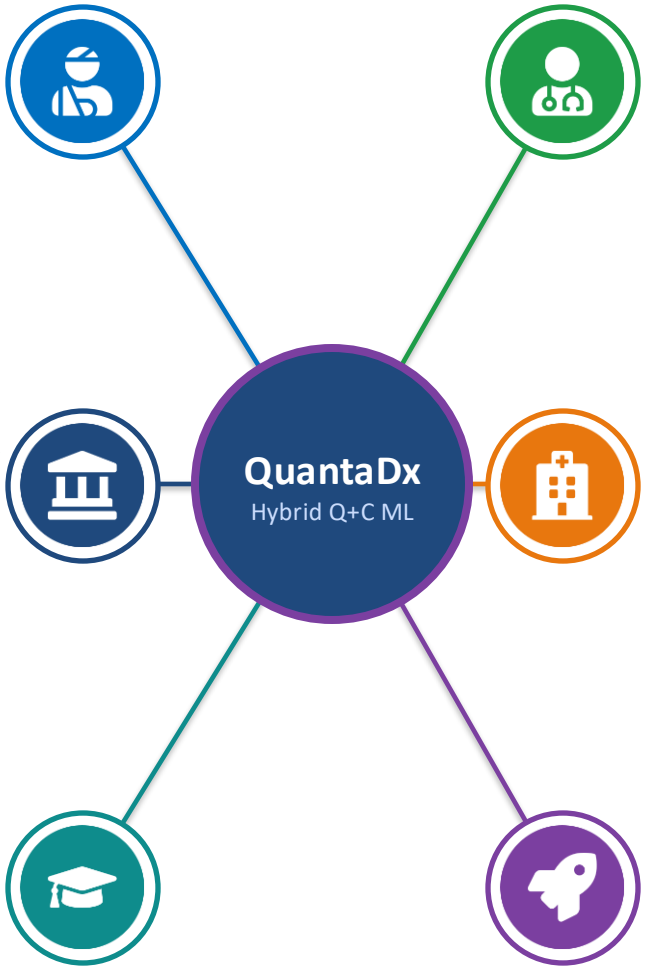
EVIDENCE BOUNDARY Results are from a public diagnostic dataset. Next: External validation (generalization) → Hospital pilot → Real-QPU validation

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IMPACT AND BENEFITS

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❖ Potential Impact on the Target Audience



Patients & Rural Communities
Earlier risk identification → timely clinical evaluation
Fewer missed cancers
Affordable screening support
Clear, simple risk report

Doctors & Clinicians
SHAP reasons on clinical features
Faster triage support
Tunable sensitivity
Second opinion in busy OPDs

Government & Public Health
Population NCD screening
ABDM & Ayushman Bharat
Data for health policy
Supports UHC & SDG 3

Hospitals & Diagnostic Labs
Automated screening workflow
HIS / EHR via REST API
Potentially lower cost per case
Scales PHC → tertiary care

Researchers & Academia
Open QML benchmark suite
Reproducible (MLflow)
Quantum-ready workforce
Testbed for new kernels

Quantum & HealthTech Industry
Real NISQ healthcare use case
National Quantum Mission
Startups, IP & jobs
Path to QPU diagnostics

❖ Benefits of the Solution

Social
Earlier detection can improve treatment opportunities; equitable care for rural & Tier-2/3 India; explainable AI builds trust

Economic
Early-stage treatment is typically less costly; fewer repeat tests; open-source software → no licence cost; startups & jobs

Environmental
Paperless digital reports; fewer **repeat scans** & patient travel; shared **cloud** compute, no new hardware

Strategic & Scientific
Advances **National Quantum Mission (₹6,003.65 Cr) [19]**; **Atmanirbhar** quantum-AI; **ABDM**-ready

ROADMAP

SDG 3 | SDG 9 | SDG 10

PHASE 1 · NOW
Current PoC: EHR / lab tables · UCI benchmarks · Hybrid Q+C · 5-fold CV

PHASE 2
External validation · genomics (small-n, high-p) · regression (severity score)

PHASE 3
Clinical translation: hospital pilot · imaging / ECG · workflow integration

PHASE 4
Quantum validation: real-QPU benchmarks · noise-aware evaluation · scaling

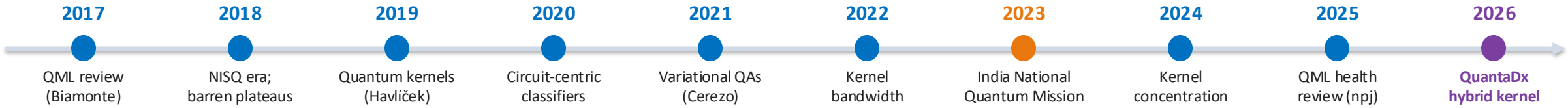
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RESEARCH AND REFERENCES

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❖ Details / Links of the Reference and Research Work

Every design choice is backed by research



Research Insights → Our Design Choices

Quantum kernels can be hard to compute classically [3, 4]
→ ZZ-map quantum kernel as a new similarity

Deep random circuits hit barren plateaus [5]
→ kernel methods + shallow ansatz

Kernels concentrate as qubits grow [11]
→ bandwidth tuning [10] + α-mix with RBF

No consistent QML advantage in health yet [15]
→ rigorous baselines + McNemar testing

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KEYWORDS: Quantum Machine Learning (QML) · Classical Front-End (CPU) + Quantum Processing Unit (QPU) · Hybrid Quantum–Classical Algorithms · Hybrid Quantum Kernel · Quantum Support Vector Machine (QSVM) · Quantum Neural Network (QNN) · Variational Quantum Classifier (VQC) · Parameterized Quantum Circuits (PQC) · Data Encoding · ZZ Feature Map · Superposition & Entanglement · NISQ / Near-Term Quantum Hardware · Quantum Simulators · Error Mitigation · Kernel Bandwidth · Kernel Concentration · Barren Plateaus · Data Cleaning · Normalization · Missing / Noisy Data · Feature Engineering · Feature Selection · Dimensionality Reduction (PCA) · Classification / Regression · Disease Probability Scores · Early Risk Stratification · Threshold Tuning · Accuracy · Sensitivity · Specificity · Computational Efficiency · Generalization · Interpretable · Explainability (SHAP / XAI) · Classical Baselines · McNemar · Early Disease Detection · Cancer · Cardiovascular · Neurological · Genomics · Medical Imaging · Electronic Health Records (EHR) · Early Detection

DATASETS: UCI Breast Cancer Wisconsin (Diagnostic) · UCI Heart Disease · Pima Diabetes · Parkinson's Voice · Chronic Kidney Disease · TCGA Genomics